

Fix the centre of force  $S$  — an immoveable point that pulls on the body. Now chop the flow of time into equal tiny slices  $\Delta t$ , and watch the body hop along the polygonal path  $ABCDEF$ : in each slice it flies dead-straight by pure inertia (Law I), then receives one sudden centripetal thump toward  $S$  at each vertex. This is Newton's *impulse approximation* — the method of first and last ratios (Lemma V, Cor. 4) applied not to figures but to *motion*. If we can show every little triangle  $S$  sweeps out has the same area, we will have proved that area swept is proportional to elapsed time — Kepler's Second Law, freed from ellipses and made true for *any* central force whatsoever.